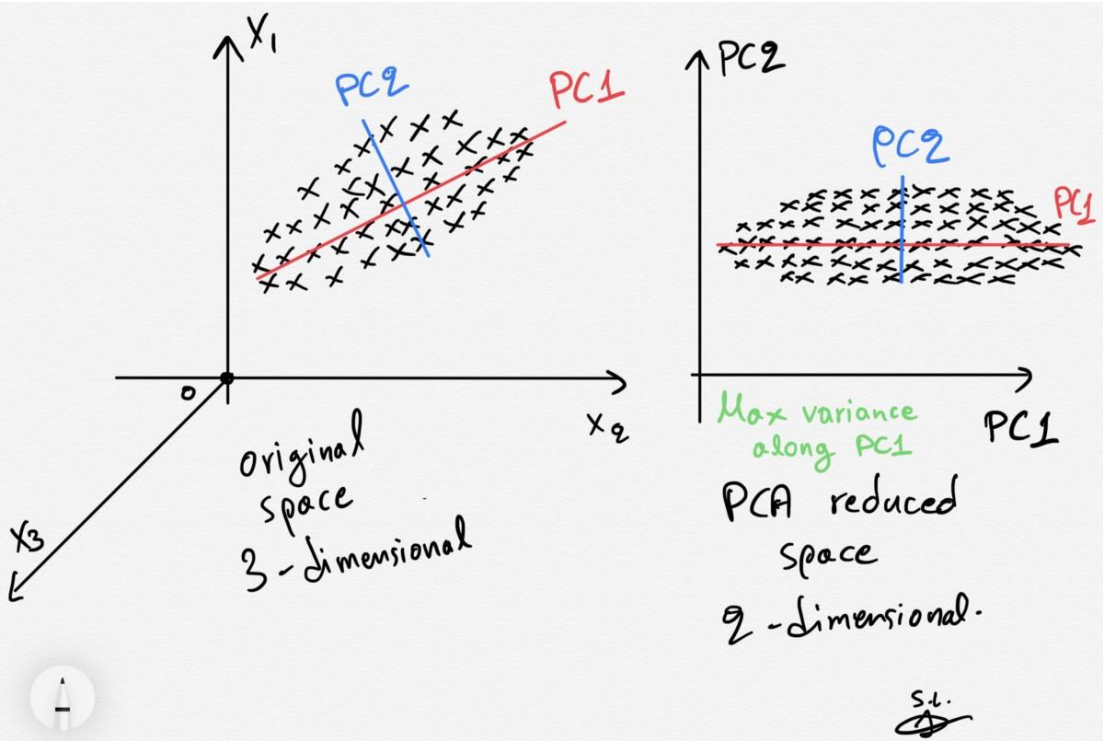
Principal Components Analysis (PCA)

http://www.cs.toronto.edu/~fleet/courses/cifarSchool09/



- Reduce dimension
- PC component describe the largest data variance

Image from https://towardsdatascience.com/pca-clearly-explained-how-when-why-to-use-it-and-feature-importance-a-guide-in-python-7c274582c37e

- Mathematically, the principal components are the eigenvectors of the covariance matrix of the original dataset.
- Because the covariance matrix is symmetric, the eigenvectors are orthogonal.
- The principal components (eigenvectors) correspond to the direction (in the original n-dimensional space) with the greatest variance in the data.

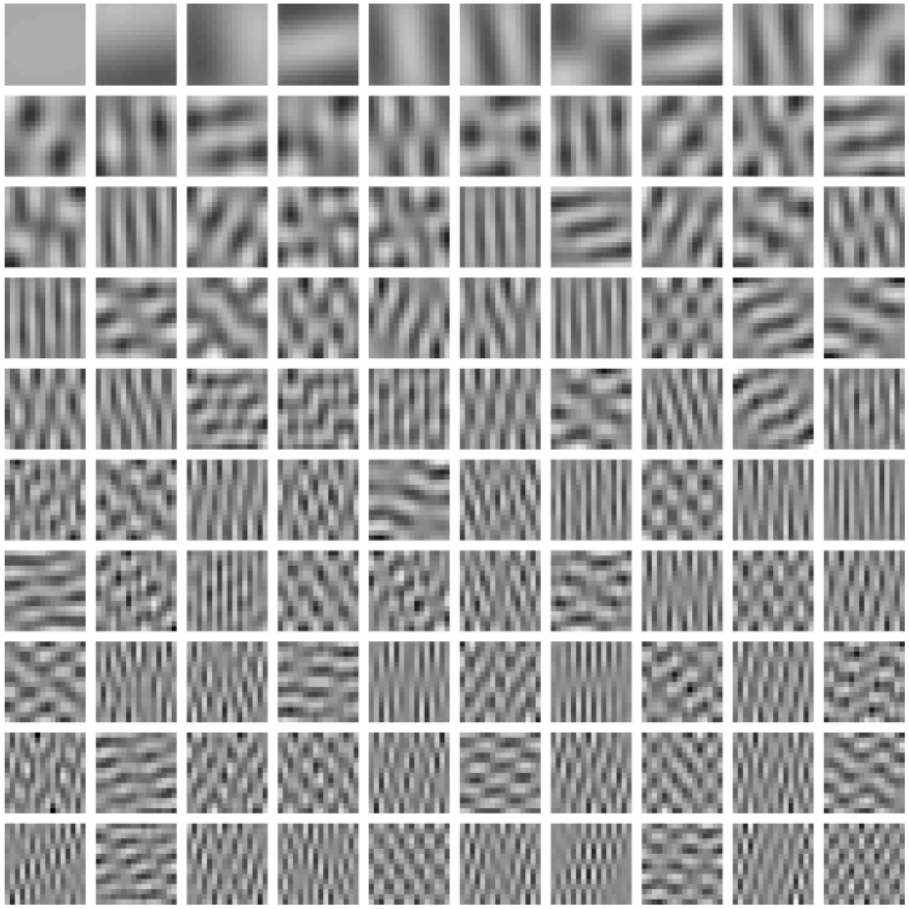
PCA

- eigen-decomposition of Σ: $\Sigma = U\Lambda U^T$
  - U: orthonormal matrix (rotation)
    $U^TU = UU^T = I$
  - Λ: diagonal matrix, $\Lambda_{ii} \geq 0$ -- eigenvalue

$$\begin{aligned} E\{U^T\vec{x}(U^T\vec{x})^T\} &= U^TE\{\vec{x}\vec{x}^T\}U \\ &= U^TU\Lambda U^TU = \Lambda \end{aligned}$$

- $s = U^Tx$, or $x = Us$, s is independent Gaussian

- principal component analysis (PCA)
  - Karhunen Loeve transform



PCA bases learned from natural images (U)